

By **Manish Gupta**

STEERING THE MACHINE: SAFEGUARDS IN THE AI ERA

AS AI BECOMES MORE ADVANCED, PROGRESS ON CONTROL AND ALIGNMENT NEEDS TO BE INTENTIONAL AND IN COLLABORATION WITH THE LOCAL ECOSYSTEM

At Google DeepMind, we treat safety not as an afterthought, but as a core scientific discipline built into foundational architectures. Guided by our Frontier Safety Framework, a set of protocols designed to prepare for and monitor extreme risks, our models undergo rigorous evaluations throughout their development lifecycle. This includes internal red-teaming to systematically uncover vulnerabilities such as prompt injection, alongside valuable external evaluations from third-party experts to ensure safety before public release.

We also build systems that respect the privacy of users. For example, MedGemma provides a privacy-first framework where sensitive health data can be processed on premises without leaving a hospital or organisation's secure ecosystem.

Responsible design extends well past launches. We incorporate active mechanisms for user feedback and bias auditing to ensure models iteratively improve accuracy, reduce hallucinations and adapt in real time. We also bake cryptographic provenance directly into model outputs. Through SynthID, we embed imper-



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